

GE Healthcare

MR750 3.0T and MR450 1.5T

Release Notes

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MR750 and MR450 Release Notes
Release Notes, English
5375780-EN TPH Rev. 2 (5/2010)

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Chapter 1: What's new

Purpose and Scope

This Release Note covers the Discovery MR750 3.0T and MR450 1.5T products. This document provides additional information that is not available in the Discovery MR750 or MR450 Operator Manuals.

The content in this chapter includes the following new content for software version 20.1_IB2:

Protocols

- Protocol Exchange
- Multi-acquisition protocols
- Shoulder protocols

Post process

- FuncTool R2*
- Missing Data Selector
- Viewer does not launch

Scan

- Calibration
- Graphic Rx
- Prescan

Miscellaneous

- 12-channel Body Array coil
- 3D FRFSE with MRCP Imaging Option
- Autoscan failure
- Exam Description text field
- Extended TPS reset
- In-Room monitor
- Planned Maintenance error message
- PROPELLER scans loses patient's Date of Birth and images
- Recon in progress
- Respiratory Trigger with 2D FSE
- Scan with invalid landmark
- Scan time
- Spectroscopy

PURE

- [PURE issues](#)

PPS

- [PPS languages](#)

User CVs

- [Half NEX Enhancement](#)
- [Whole Volume Excitation](#)

Protocols

- **Protocol Exchange:** When using Protocol Exchange Export mode, all pediatric protocols (site or GE library) are now listed. Therefore, all pediatric protocols can be exported.
- **Multi-acquisition protocols:** The Acqs before Pause value is no longer lost when saving and reloading a protocol.
- **Shoulder protocols:** When acquiring PURE shoulder images with the 8-channel shoulder coil, be sure that the calibration scan is acquired with frequency direction R/L to match the scan frequency direction. A Protocol Notes for Adult->upper extremities -> Shoulder -> Calibration sequence has added the following note: "Ensure that the calibration frequency direction is R/L".

Post process

- **FuncTool R2*:** The average value for an ROI is now displayed in units of Hz and not kHz.
- **Missing Data Selector:** The following issue has been resolved and the scenario no longer results in no Data Selector displayed on the Image Management screen.
 1. Select an exam from the patient list that has number symbol (#) in the patient name field.
 2. From the Sessions Apps list, click Multi-Exam Viewer.
 3. Switch between review session tab and the Image Management desktop and the Data Selector disappears on the review session tab.
- **Viewer does not launch:** Select an exam from the Patient List and click **Viewer** from the Session Apps list and the Viewer does not launch. This issue is resolved.

Scan

Calibration

- **Calibration data:** PROPELLER scans no longer lose the calibration data when you switch between PROPELLER and other non-PROPELLER series. Therefore the calibration data can be used for all scans within a series as long as all other calibration considerations are followed.
- The following **calibration scan and TPS** reset issue has been resolved.
 - If you have acquired calibration scans from multiple coil modes, execute a TPS reset, and then acquire scans in the calibration anatomical area, you may see image quality issues (banding and alias artifacts). Acquire the calibration scans again and then acquire the ASSET scans again.

Graphic Rx

- **Graphic Rx:** During a graphic prescription, there were occasions when the system crashed due to the graphic prescription geometry code. These issues are resolved.
- **Graphic Rx toolbar**
 - Occasionally the Graphic Rx toolbar disappeared from the screen during scan prescription. It could not be redisplayed by clicking the Graphic Rx icon. This issue is resolved.
 - The Graphic Rx toolbar also did not appear after a Real Time scan was done and another scan was in the Setup mode. This issue has also been resolved.

Prescan

- **Auto Prescan error message:** On 3.0T systems, Auto Prescan no longer stops due to the "signal too small" error.
- **Auto Prescan failure:** Auto prescan no longer fails after you press the Move to Scan button.

Miscellaneous

- **Exam Description text field:** The Exam Description text field on the WorkList Manager Exam screen can now be edited on systems with HIS/RIS configuration.
- **Scan with invalid landmark:** You can no longer download a series with an invalid landmark. A message prompt appears indicating that a landmark is required to proceed to scan. The following issue has been resolved.
 - Define a pending landmark using IntelliTouch landmark strip and then do not accept it.
 - Start the scan and the images do not appear in Auto Viewer.
- **Scan time:** The scan time accurately counts down for all scan prescriptions.
- **3D FRFSE with MRCP Imaging Option:** When Auto Transfer by Series is selected on, only the collapsed images were transferred. This issue has been resolved and now all images are transferred to the selected remote node once scan and reconstruction is completed.
- **Spectroscopy:** 2D CSI scan data no longer fails to reconstruct.
- **Respiratory Trigger with 2D FSE:** The maximum TR for a respiratory triggered scan has been changed from 15 to 50 seconds. This should accommodate patients with low respiratory rates.
- **In-Room monitor:** On occasion, the trackball loses contact with the In-room monitor. This issue is resolved.
- **Recon in progress issue:** On occasion, when Shim Volume was applied to a series and Shim value was set to **Auto**, the next series could not begin until reconstruction completed on the Auto Shim series. This issue is resolved.
- **PROPELLER scans loses patient's Date of Birth and images:** Based on the method used to pull the patient's DOB from a HIS/RIS system, the date of birth and patient age fields were left blank on non-PROPELLER images and the PROPELLER images were not listed in the Patient List, thus appearing to be lost. This issue is resolved.
- **12-channel Body Array coil:** The Body_Upper mode of the coil is not compatible with the PURE/ASSET Calibration scan, due to the inability to reduce the calibration FOV. This issue is resolved.
- **Extended TPS reset:** After system start-up, an Extended TPS reset can occur. This issue is resolved.
- **Autoscan failure:** On a 3.0T system, if Autoscan is selected and the table needs to move more than 5cm, Autoscan fails, the currently prescribed series in the RXD state moves to New state, and the Scan and End Exam buttons are not available. This issue is resolved.
- **Planned Maintenance error message:** An error log PM message to "call service immediately," prevents scanning. This issue is resolved.

PURE

- The following **PROPELLER and PURE** issue has been resolved.
 1. You prescribe a calibration scan with a particular coil.
 2. Acquire a PROPELLER and PURE scan within the calibrated area.
 3. Prescribe slices outside of the calibration scan with the same coil.
 4. Prescribe and acquire another calibration scan in the new anatomical area.
 5. Acquire another PROPELLER with PURE scan. The system uses the first calibration file and not the most recently acquired calibration file.
 6. End the exam and re-prescribe the calibration scan in the desired area and then acquire the PROPELLER with PURE scan. This problem does not occur if you switch coils between calibration scans. It only occurs when both calibration scans are acquired with the same coil or coils within a phased array coil.
 7. To avoid this scenario, make sure you have a large enough calibration scan coverage to accommodate all potential scan locations.
- The following **PURE** issue has been resolved.
 - Some PURE scans were over-ranged resulting in the inability to appropriately adjust the Window width and level to demonstrate detail. The software fix changes the typical image intensity of PURE scans. After installation of the software, please check the settings of post-processing applications to ensure continuity of results.

PPS

- The following **PPS** issue has been resolved.
 - For sites that use languages other than English on the scanner and use Performed Procedure Step (PPS) to tell the RIS or PACS when an exam has been performed, the PPS feature may be non-functional when using a RIS or PACS that requires character set description information to be passed. In particular the tag (0008,0005) for character set information (such as 'ISO_IR 100') is not sent as part of the PPS communications which may be required.

User CVs

- There is a new **Half NEX Enhancement** User CV for 2D FSE scans.
- **Whole Volume Excitation User CV** content has been updated.

Chapter 2: New considerations

The content in this chapter includes the following considerations:

- [Protocols](#)
- [Post Process BrainStat images](#)
- [MPR fails to complete from InLine Viewer](#)

Scan issues

- [Localizer](#)
- [3DFRFSE-XL with MRCP](#)
- [PROPELLER and auto scan](#)
- [BRAVO](#)
- [Linked series](#)
- [Sagittal VIBRANT-Flex with Dynaplan \(Multiphase\) reconstruction](#)

Protocols

- If you are building or moving protocols from an HDx based MR system to your Discovery system, when a setup Protocol routine is run during a system start-up, HDx protocols display in an Invalid Protocols folder. This is expected behavior. Discovery systems do not support HDx based protocols.

Post process

- **FuncTool BrainStat:** Please disregard the BrainStat AIF information that is in the FuncTool section. It is currently unavailable.
- **BrainStat functional maps:** Functional maps created in FuncTool BrainStat on the MR console and transferred to an AW console (4.5) cannot be opened with the AW Volume Viewer. To resolve this issue view the functional maps from the MR console Volume Viewer.
 - Note that functional maps created on an AW system that has FuncTool 6.x or earlier, can be opened with Volume Viewer on AW.
- If you have an Auto MIP post process task added as part of the series and the PSD adds images such as MAG and Phase images to the data base (for example 3D Inhance Velocity or 3D TOF single slab series), the Auto MIP post process task can result in an error. If this condition occurs, use manual MIP.
- You cannot launch Flow Analysis (selected from the Session Apps list in the Image Management work area) if the selected exam has a GSPS series in the patient list. To avoid this problem, do not create a GSPS series until after you have launched the Flow Analysis application.

MPR fails to complete from InLine Viewer

If a prescribed MPR task does not complete, the task in the Workflow Manager is labeled ERR (error). Navigate to the Image Management work area, select the patient's exam/series from the Patient List, and from the Session Apps list click **Reformat** to manually complete the MPR.

Scan

- If you prescribe another series while the localizer is being acquired, it results in an error message that indicates you cannot download another sequence. Once the localizer scan is completed, the localizer Worklist Manager displays the task state as Done. Even though the state displays as Done, you will not be able to download any other series, nor can you end the scan session. You must reboot the system. Therefore, do not prescribe another series until the localizer scan is finished.

3DFRFSE-XL with MRCP

- When acquiring a 3D FRFSE-XL with MRCP imaging Option turned on, after the acquisition is finished, the scan clock continues to count down. This is normal behavior. The 3D FRFSE-XL with MRCP breath hold scan uses a high RF power mode that allows a very short scan time. Therefore, an additional patient cool down time is required to prevent the start of another acquisition too soon. Once the clock is a zero, you can begin another acquisition.

PROPELLER and auto scan

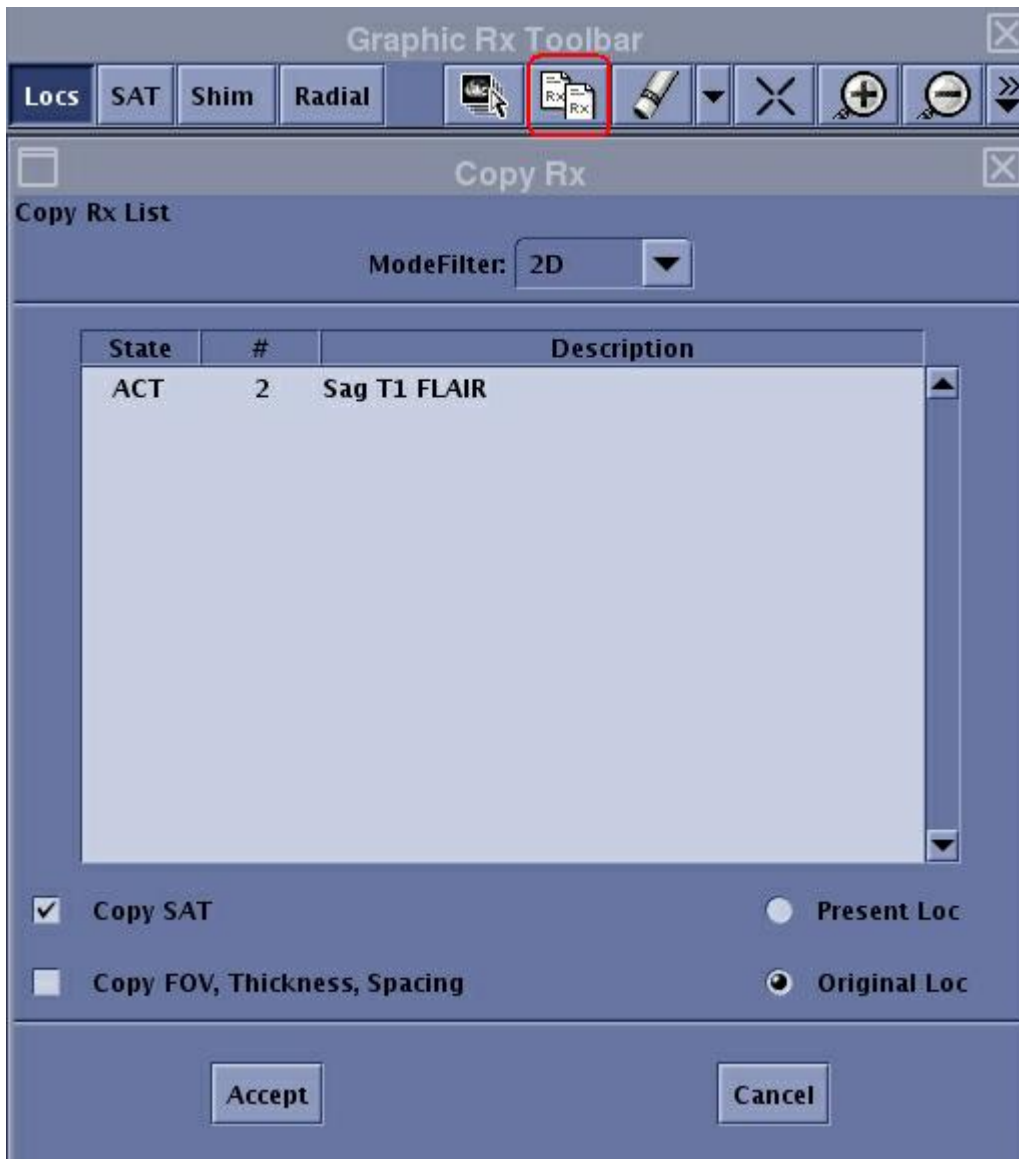
- The following conditions result in very noisy PROPELLER images.
 1. Start an exam.
 2. In the Workflow Manager, have a localizer and two PROPELLER series, for example a T2 PROPELLER and T2 FLAIR PROPELLER.
 3. Save the localizer and select **Auto Scan**.
 4. Select the T2 PROPELLER and click **Setup**, prescribe the slices and click **Save Rx**.
 5. Wait for Auto Scan to begin the T2 PROPELLER prescan.
 6. From the Workflow Manager, select the T2 FLAIR PROPELLER series and click **Setup**.
 7. While the system is executing prescan on the T2 PROPELLER series, prescribe the slices on the T2 FLAIR PROPELLER series and click **Save Rx**.

The T2 PROPELLER images are very noisy. To avoid this problem, Save Rx for both PROPELLER scans before you click Auto Scan. Following this workflow, the PROPELLER scan image quality is as expected.

BRAVO

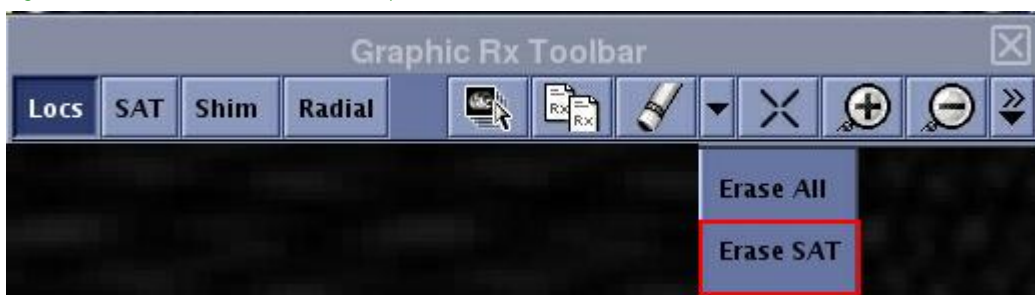
SAT pulses are not compatible with BRAVO. If you erase the existing graphic prescription on a BRAVO series, and then from the Graphic Rx toolbar, click **Copy Rx** icon to copy the slice locations from another series and Copy SAT is turned on, you must turn off Copy SAT before you click **Accept**.

Figure 2-1: Copy Rx screen with Copy SAT selected. De-select it before you click **Accept**.



If you do not turn off Copy SAT and click Accept, an error message displays: "SAT is not compatible with IR Prep Option in fast 3D gradient echo". At this point, from the Graphic Rx toolbar, click the Erase icon and Erase SAT from pull-down menu.

Figure 2-2: Click **Erase SAT** from the Erase pull-down menu



Linked series

When multiple series are linked together, one or more of those linked series may not change from the **New** state to the **RXD** state. Series in the **New** state may not allow Graphic RX and may have scan parameters

that are not editable. Rebuild the corrupted series using a GE protocol or template and re-link the series. This issue is resolved.

Sagittal VIBRANT-Flex with Dynaplan (Multiphase) reconstruction

Slices at the beginning and end of a multi-phase, Sagittal VIBRANT-Flex scan, may fail to reconstruct due to memory limitation. Consider the following resolutions to avoid this issue:

- Select **Variable Delay** from the multi-phase tab, which may limit the maximum number of phases that can be scanned.
- Prescribe a multi-phase Axial VIBRANT-Flex scan, which allows the use of Variable Delay.
- Prescribe a standard Sagittal or Axial VIBRANT scan.

Chapter 3: User CVs

The content in this chapter includes the following two User CVs:

[Half NEX Enhancement](#)

[Whole Volume Excitation](#)

Half NEX Enhancement User CV

Use **Half NEX Enhancement** to reduce ghosting artifact that potentially can appear on FSE half NEX scans. It may also reduce ghosting for one NEX scans with NPW option.

- Select **1** to turn it on. The default is off (0).

Figure 3-1: User CV on/off illustrating ghosting artifact reduced

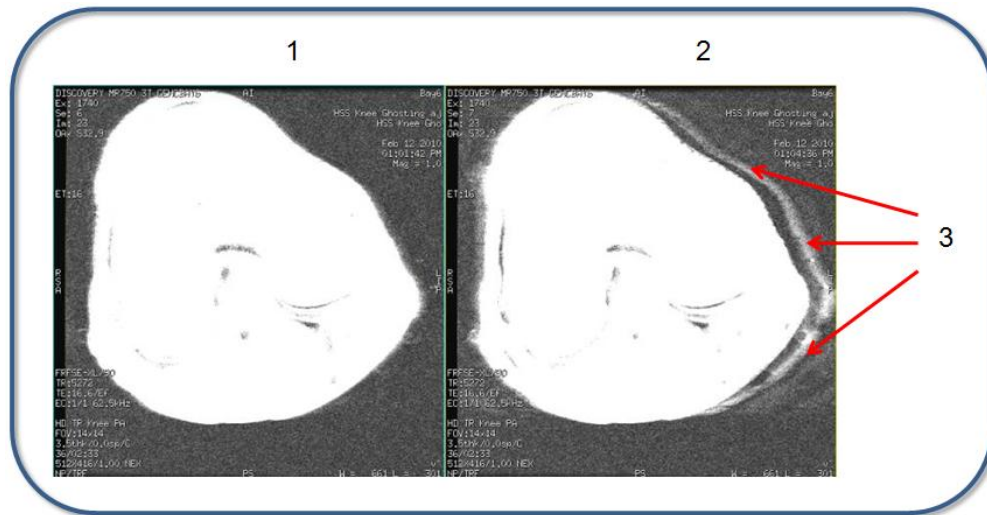


Table 3-1: Image legend

#	Description
1	Half NEX Enhancement turned on with minimized ghosting
2	Half NEX Enhancement turned off, which demonstrates ghosting
3	Ghosting artifact

Whole Volume Excitation

The **Whole Volume Excitation** can be selected from multiple PSDs.

Whole Volume Excitation with Cube

The Whole Volume Excitation User CV is used to reduce the shading or banding artifacts observed in some cube images (acquired or reformatted).

- Be sure to cover the whole anatomy that is scanned, otherwise aliasing artifacts will be observed.
- This CV is not compatible with the Flow Compensation imaging option. This CV is not compatible with Axial and Oblique Axial planes.
- Whole Volume Excitation 1 = **On**: The Excitation RF, Readout Dephaser Gradient & Slice Rephaser Gradient are applied in a unique manner that results in reducing shading or banding artifact.
- Whole Volume Excitation 0 = **Off**: Turns off Whole Volume Excitation.

Whole Volume Excitation with 3D Fast GRE applications and PSDs

This User CV is applicable for Sagittal and Coronal scans with many 3D Fast Gradient Echo Applications and PSDs such as LAVA-Flex, LAVA, VIBRANT-Flex, VIBRANT, BRAVO, 3D FSPGR/FGRE, 3D Fast TOF GRE/SPGR, and TRICKS.

- Whole Volume Excitation 1 = **On**: Very short duration hard RF pulse is used without slab selective gradient pulse, which results in shorter TE/TR and improves signal uniformity at edge slices but introduces wrap in the slice direction. The slab must be large enough to include all anatomy to avoid wrap in slice direction. It is recommended that you not use it when FTMRA is selected.
- Whole Volume Excitation 0 = **Off**: If you cannot cover the anatomy that is scanned in the slice direction, turn off Whole Volume Excitation. An exception is (other than for LAVA-Flex, VIBRANT-Flex or if slice acceleration is used), if the wrap artifact will not fall within the region of interest; in this case you could turn it On.